

Meridian Pipeline Intelligence

Everything you need to explain the web application clearly, in plain English, for a 60-minute client walkthrough — what it does, how it works, what the data revealed, and why it can be trusted.

Scenario: Meridian Systems sales-intelligence engagement · Data as of May 2, 2026 (Q1 final, Q2 day 32 of 91) · Prepared by Sai Saran

How to use this document

Each chapter has a suggested time in its heading; the total is roughly 60 minutes. Blue boxes tell you **what to show in the app** at that moment. Green boxes give you a **key line** worth saying close to word-for-word. You do not need to cover everything — chapters 1–7 are the core; 8–12 are depth you can pull in when questions go there.

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1 • The client's problem

≈ 5 min

Meridian Systems sells enterprise software. Their sales organization is large and spread out, and the information about how sales are going lives in several disconnected places — dashboards that don't talk to each other, and analysts who prepare reports on request. The data exists. Getting an answer out of it is the problem.

When a senior leader asks something simple — *"How is the Enterprise segment tracking against quota this quarter?"* — the honest answer today is: wait. Wait for an analyst to pull the numbers, or bounce between three dashboards and reconcile them yourself. Decisions slow down. Eventually, leaders stop asking questions altogether, and start running the business on gut feel. That is the real cost: not wrong answers, but **unasked questions**.

Eighteen months ago they tried to fix this with an AI reporting tool. It made up numbers — the industry word is "hallucinated" — and mixed up customer segments. The Chief Commercial Officer caught it doing so in a meeting, in front of people. There is no formal ban on AI at Meridian, but the skepticism is real, and it starts at the very top.

KEY LINE — say it in these words

The bar here was never "does it work." The bar is "can we trust it." Every design decision in what I'm about to show you was made to clear that second bar.

So the assignment was: build something that lets a sales leader ask a plain-English question about the pipeline and get back three things — a **direct answer** (not a hedge), the **actual data behind the answer** (not "based on your data"), and a **clear flag** whenever the answer involves incomplete data, ambiguity, or an assumption.

One framing worth stating up front: the product I'll show you competes with *distrust*, not with dashboards. Its job is to be the first AI-flavored tool this CCO doesn't have to double-check — and it earns that by double-checking itself, visibly, on every answer.

2 · What we built — the tour

≈ 6 min

What we built is a small web application. It looks and feels like a chat: a leader types a question in ordinary English, and gets back an answer. But under the surface it behaves nothing like a chatbot — and that difference is the whole point. Walk the client through the screen top to bottom.

SHOW IN THE APP

Open the app full-screen. Don't ask anything yet — tour the resting state first. It already tells the story of the quarter before a single question is asked.

The header and the four big numbers

At the top: the product name and one sentence of promise — *"numbers are computed, never generated."* Below it, four cards that animate up when the page loads. These are the state of the quarter at a glance:

- **Q2 quota — \$6,200,000.** The total sales target for the ten salespeople this quarter.
- **Booked, Closed-Won — \$518,000.** Deals actually won so far. Note the small print: *"strictly new \$145,000 (F4)"*. The tool is already disclosing a data problem — we'll get to it — where most of that \$518K was arguably counted last quarter too.
- **Open pipeline — \$5,198,000.** The value of deals still being worked that are expected to close this quarter. The card explicitly says this is *not booked revenue* — it's potential, not money.
- **Uncovered quota — \$484,000.** The most important number on the screen: even if Meridian won *every single open deal*, they would still fall \$484,000 short of the quarter's target. No optimism required to compute that — it's subtraction.

KEY LINE — say it in these words

Before anyone asks a single question, the tool has already told you the quarter's headline: the math doesn't close. Even a perfect win rate leaves you \$484,000 short — so the conversation isn't "will we win our deals," it's "we need to create pipeline, now."

The left sidebar — the tool's conscience

The sidebar is where the tool is honest about itself:

- **Data status:** "Data validated · 5 reconciliations passed." Every time the app starts, it re-checks its own math five different ways (totals computed deal-by-deal must equal totals computed per-person and per-segment, and so on). If any check fails, the app refuses to show numbers at all.
- **AI assist:** a switch showing whether the language model is on, and a one-line contract: the model helps with wording and routing, but *numbers are always computed, never generated*. Flip the switch off live if you want — the app keeps working.
- **Known data issues:** five findings, labeled F1 through F5, that the tool discovered in Meridian's data *by itself*, automatically, when it loaded the files. This list is Chapter 4 — it's the most valuable output of the whole engagement.

- **Methodology:** every metric definition, every threshold, every assumption — written down where any skeptic can read it.

The suggested questions and the chat

Five capsule buttons — Enterprise vs quota, Reps at risk, Q2 vs Q1 pace, Regional view, Data quality — cover the questions leadership asks most, one click each. Below them, a free-text box for anything else. Each answer comes back as a card containing, in order: what the question was interpreted as, a color-coded trust badge, the answer in one or two plain sentences, the key numbers as tiles, a chart when one helps, any data-quality warnings, and three tabs — **Source data** (the actual rows behind the answer), **Calculation** (the exact arithmetic), and **Assumptions**.

KEY LINE — say it in these words

Every answer in this tool comes with receipts. If you don't believe a number, you don't have to argue with the tool — you click one tab and read the exact rows it came from.

3 • The data we received

≈ 5 min

Meridian shared four spreadsheet files — two snapshots of the same sales system, taken at two moments in time:

FILE	WHAT IT IS	ROWS
Q1 deals	Every sales opportunity as recorded when the books closed on Q1 (January–March 2026). Declared final .	88 deals
Q1 reps	The ten salespeople, their territories, and their Q1 targets ("quotas").	10 reps
Q2 deals	The same deal table re-photographed on May 2, 2026 — 32 days into Q2. Contains all the Q1 deals <i>plus</i> new Q2 activity.	92 deals
Q2 reps	The same ten people, now with Q2 targets added.	10 reps

What a deal record contains, in business terms

- **Deal ID** — a serial number like OPP-042. Supposed to be a stable identity for one opportunity. (Chapter 4 will show it isn't.)
- **Account** — the customer company being sold to.
- **Segment** — the customer's size class: Enterprise (biggest), Mid-Market, or SMB (small business).
- **Region** — West, Central, Northeast, or Southeast.
- **Rep** — which salesperson owns the deal.
- **Stage** — how far along the sale is: Prospecting → Discovery → Proposal → Negotiation, ending in either **Closed Won** (we got the business) or **Closed Lost** (we didn't).
- **Deal value** — the dollar size.
- **Close date** — when it closed, or for open deals, when the rep *expects* it to close. Important nuance: for open deals this is a forecast, not a fact — 44 of the 88 Q1 deals slipped their dates between the two snapshots.

Three ground rules we set before computing anything

1. **A deal belongs to a quarter by its close date** — never by which file it appears in, because the Q2 file also contains all of Q1's history.
2. **Q1 is final means Q1 is final.** All Q1 reporting comes from the Q1 file. When the Q2 snapshot quietly disagrees with closed Q1 history — and it does, repeatedly — we flag the disagreement rather than silently adopting either version.
3. **Won money and hoped-for money never mix.** "Attainment" counts only Closed-Won deals. Open pipeline is always shown separately, clearly labeled as not-yet-revenue.

KEY LINE — say it in these words

These three rules sound obvious. The interesting thing is that the previous AI tool failed precisely because it had no rules like them — it blended files, blended quarters, and blended hope with revenue. Discipline about boring

rules is where trust actually comes from.

4 • The five data findings

≈ 8 min

Before building any answers, we compared the two snapshots record by record — the same forensic pass a careful analyst would do. Five distinct problems surfaced. Together they touch **28 of the 92 deals** in the current snapshot. The app detects all five automatically every time it loads: nothing is hardcoded, so if Meridian fixes the source data, the flags disappear on their own.

SHOW IN THE APP

Click the **Data quality** chip. This entire chapter is on screen: the five findings as tiles, a red chart of dollar impact, and — in the Source data tab — every affected row, tagged, next to its Q1 version. Note this is the one answer with a **green** badge: the report itself is exact.

F1 — Recycled deal IDs (13 deals • \$1.97M • 38% of open pipeline)

Thirteen deal IDs — OPP-076 through OPP-088 — refer to **completely different companies** in the two snapshots. In Q1, OPP-076 was "Cascade Financial," a small-business deal, already won, worth \$112,000. In Q2 the same ID is "Evergreen Capital," a Mid-Market prospect in early Discovery, worth \$250,000. Someone deleted old records and reused their ID numbers.

Why it matters: any report that compares "the same deal" across time using these IDs produces nonsense. And it's not a corner case — \$1,970,000, roughly 38% of the entire open pipeline, sits on these reused IDs. **Our policy:** the current rows are treated as the current system of record (they're real opportunities), but they are banned from any deal-history comparison, and every answer they touch carries a flag.

F2 — Closed deals that came back to life (7 deals • \$576K)

Seven deals that were **closed** in the final Q1 book — some won, some lost — show up in the Q2 system as open again, or with a different outcome. For example, OPP-020 was recorded as Won in Q1; in Q2 it's back in Negotiation. **Why it matters:** "final" is supposed to mean final; a closed quarter's revenue shouldn't quietly change. Also, \$466,000 of the current pipeline is deals like these — and \$253,000 of that was already counted as Q1 revenue, so there is overlap between last quarter's results and this quarter's hopes. **Our policy:** Q1 reporting follows the closed Q1 book; the deals count in Q2 only as open pipeline; the overlap is flagged wherever it appears.

F3 — Overdue open deals (2 deals • \$87K)

Two open deals (OPP-066, OPP-075) are past the close dates their reps forecast. Minor hygiene, kept in the pipeline, flagged.

F4 — The big one: revenue counted in both quarters (3 deals • \$373K • 72% of "Q2 booked")

Three deals — OPP-042, OPP-060, OPP-074 — are recorded as **Closed Won in both snapshots**, but their close dates were moved: won in late March according to the final Q1 book, won in early April according to the Q2 system. The same revenue is counted in Q1's results *and* in Q2's results.

Why it matters more than everything else: of the \$518,000 the Q2 system calls "booked this quarter," \$373,000 — seventy-two percent — is re-dated Q1 revenue. Truly **new** Q2 sales are \$145,000. And for the Enterprise segment it's starker: its single Q2 win, \$165,000, is one of these three deals — strictly new Enterprise revenue this quarter is **zero**. This is also an audit and compensation concern, not just an analytics one: moving revenue across a closed quarter boundary is the kind of thing finance teams get asked hard questions about.

Our policy: we don't rewrite the client's books unilaterally. Every affected answer shows both figures — the system-of-record number and the strictly-new number — and any answer where re-dated revenue dominates **downgrades its own trust badge to red**.

KEY LINE — say it in these words

I want to be transparent about one thing: my first working version of this tool passed every test I'd written and still missed this finding. I caught it in an adversarial audit pass afterward — deliberately trying to refute my own outputs against the raw files. That experience is built into the product now: F4 detection is automatic, and it's why I'll never tell you this tool can't be wrong. I'll tell you it's built so that being wrong is detectable, attributable, and cheap to fix.

F5 — History edited after the books closed (3 deals · \$260K)

Three closed deals had their value or date quietly edited between snapshots without their outcome changing. The clearest: OPP-021, closed and won in Q1 at \$155,000 — the Q2 system now says the same closed deal was worth \$120,000. A \$35,000 write-down of history, unexplained. **Our policy:** Q1 reporting follows the Q1 file; the edits are surfaced, not silently adopted. **Why it matters:** anyone recomputing Q1 from the current system gets different Q1 totals than the closed books — a guaranteed "whose-number-is-right" meeting unless it's surfaced first.

The one-sentence summary of this chapter: the question "what's wrong with my data?" turned out to be as valuable as any sales question — which is why we made it a first-class, one-click question in the tool itself.

5 • How the app answers a question

≈ 6 min

This is the chapter where you explain the architecture — in plain English, because the idea is genuinely simple. The last AI tool failed because one system did two jobs: it both *understood the question* and *produced the numbers*. When the same model does both, every answer is a potential fabrication. We separated the jobs permanently.

The restaurant analogy

Think of it as a restaurant. The AI is the **waiter**: it listens to what you ask for, figures out which item on the menu you mean, and later describes the finished dish nicely. But the waiter never cooks. The **kitchen** is ordinary, boring, checkable computer code — the same arithmetic an analyst would do in a spreadsheet, just automated and double-checked. And the menu is short on purpose: **eight dishes**, each one prepared, tasted, and audited before it was ever served.

The five steps, every single time

1. **Your question arrives.** "How is Enterprise tracking against quota?"
2. **Routing — choosing from the menu.** First, simple transparent keyword rules try to match the question to one of the eight analyses. If they can't, the AI is allowed one narrow job: pick which of the eight fits — it literally cannot answer anything else. If nothing fits, the tool **refuses**, politely and specifically. It never guesses.
3. **Computation — the kitchen.** Ordinary code filters the validated data and does the arithmetic. No AI is anywhere in this step. The same question always produces the same numbers.
4. **Narration with a veto.** The AI may rewrite the computed result into a friendly sentence or two. Then a **validator** — plain code — extracts every number from that sentence and checks each one against the numbers the kitchen actually produced. If even one number doesn't match, the whole AI sentence is thrown away and the tool shows its own plain wording instead — and tells you it did so. A made-up number *cannot reach the screen*.
5. **The answer ships with receipts:** source rows, the formula, warnings, assumptions, and a trust badge.

SHOW IN THE APP

Type: **Are we going to make the number this quarter?** Point at the small gray line above the answer: "LLM classification, validated against the approved intent catalog" — the waiter chose from the menu. Then point at "AI summary · N numbers verified against computed results" — the veto in action.

KEY LINE — say it in these words

*Here's the trade we made, stated honestly: this tool can answer eight kinds of questions, not eight thousand. In exchange, the worst thing that can ever happen is a **visible** mistake — it picks the wrong menu item and you see that immediately at the top of the answer. The worst case of the unrestricted approach is an **invisible** mistake: a plausible, confident, wrong number. This client has already lived that. We chose the failure mode you can see.*

And the menu grows. Each new question type gets built, tested, and audited before it ever answers a leader — so capability expands week by week without ever re-opening the trust question.

6 • The eight analyses, one by one

≈ 12 min

This is the heart of the demo. For each analysis: what it answers, how it's calculated in words, what it says about Meridian today, and what to show.

6.1 Enterprise vs quota ● Insufficient data

The question: "How is the Enterprise segment tracking against quota this quarter?" — the CCO's own example question.

The calculation, in words: add up every Enterprise deal won between April 1 and May 2 (\$165,000 — it's one single deal, Sunridge Systems); divide by the combined Q2 targets of the four Enterprise reps (\$3,650,000). Result: **4.5% attainment** with 35% of the quarter gone. Then look at the open Enterprise pipeline: \$3,600,000, which covers the remaining gap 1.03 times — below the 1.28× that history says is needed (explained in 6.2).

But the badge is red. The tool downgrades its own answer, because that entire \$165,000 is F4 re-dated revenue — a deal the closed Q1 book says was won on March 20. Strictly new Enterprise revenue this quarter: **\$0**. On top of that, \$1,475,000 of the Enterprise pipeline sits on recycled IDs (F1) and \$305,000 is reopened previously-closed deals (F2).

SHOW IN THE APP

Click **Enterprise vs quota**. Open the Source data tab and show that the numerator is one row. Say: "when the answer is one deal, the tool shows you the deal."

KEY LINE — say it in these words

A normal dashboard would show 4.5% and move on. This tool says: 4.5% is arithmetically correct and strategically misleading — and here's exactly why, down to the row.

6.2 Reps at risk ● Verified w/ assumptions

The question: "Which reps are at risk of missing Q2?"

How risk is decided — three visible signals, no black box:

- **Pace** — what share of their target has each rep won, compared to how much of the quarter has passed (35%)?
- **Coverage** — is there even enough open pipeline to close their remaining gap? Coverage below 1.0× means the target is *arithmetically unreachable* — even winning everything wouldn't get there. And 1.0× is generous: last quarter Meridian actually won 77.9 cents of every decided dollar, so realistically you need about **1.28×** coverage ($1 \div 0.779$). That threshold comes from *their own history*, not an industry rule of thumb.
- **Record reliability** — if more than half a rep's pipeline sits on recycled IDs (F1), the tool refuses to judge the person at all.

The verdicts today: **At risk** — Kevin Marsh, Lisa Park, Ryan Cole: zero booked and coverage below 1.0×; the math is short no matter what. **Unclassifiable** — Priya Patel, Danielle Torres, Aisha Williams: 54–67% of their

pipeline is on corrupted records — that's a data problem, not a performance problem. Everyone else is "Watch." The standout performer is Tom Bradley at 56.3% of target — though the tool notes 71% of his booked figure is F4 re-dated revenue.

The honest caveat the tool prints itself: it's day 32, and seven of ten reps have booked nothing — but Q1 looked identical at day 32 (21% booked) and still finished at 102%. So "At risk" means "needs pipeline scrutiny now," not "will fail."

SHOW IN THE APP

Click **Reps at risk**. Show the coverage chart with the two dashed lines: 1.0x (impossible below this) and 1.28x (break-even at their real win rate). Then point at the three gray "Insufficient data" rows.

KEY LINE — say it in these words

This tool will tell you a rep is behind. It will refuse to tell you a rep is failing when the real problem is that their records are corrupted. That distinction protects careers — and it's exactly the judgment a hallucinating tool can't make.

6.3 Q2 vs Q1 at the same point ● Verified w/ assumptions

The question: "How does Q2 attainment compare to where we were at the same point in Q1?"

The fairness problem, solved simply: "same point" is defined as **equal days elapsed** — today is day 32 of Q2, so we compare against Q1 through *its* day 32, which was February 1. We checked the residual calendar effects: Q2's window actually has one *more* business day, so the alignment slightly favors Q2 — meaning the gap you're about to see is real, not an artifact.

The result: Q2 has booked \$518,000 (8.4% of quota) versus \$1,260,000 (21.3%) at the same point in Q1 — **about 41% of Q1's pace**. On a strictly-new basis (removing the \$373K of F4 re-dated revenue) it's \$145,000, and the gap widens further. Context that keeps it honest: Q1 was also slow early and finished at 102% (\$6,041,000) — quarters here are back-loaded. But pace this far behind, plus pipeline that can't cover the gap, is why the tool's company-level conclusion is "create pipeline now."

SHOW IN THE APP

Click **Q2 vs Q1 pace**. Show the two-bar chart, then the "Strictly-new Q2 wins: \$145,000" tile.

6.4 Individual rep performance

Ask it live: "How is Priya Patel doing?" — \$0 booked of a \$1,000,000 target, \$740,000 pipeline covering 0.74x, and a **red badge**: 61% of her pipeline is on recycled IDs, so the tool declines to draw performance conclusions and names the fix — correct her seven affected records, then re-ask. Also try Tom Bradley for the contrast: strong numbers, with the F4 asterisk shown plainly.

6.5 Segment performance

The same treatment as 6.1 for any segment. Useful contrast: **Mid-Market** is the healthiest segment — \$353,000 booked (17.3%) — though \$208K of it is re-dated; **SMB** has booked nothing, but is also the only segment whose

pipeline is completely free of recycled IDs, so its answers carry cleaner badges.

6.6 Regional view

Revenue and pipeline broken down West / Central / Northeast / Southeast. One honest wrinkle the tool discloses: revenue is credited to the *deal's* region, but targets belong to the *rep's home* region — and some deals are sold outside their owner's home region. So regional attainment is labeled directional, with the mismatched deals listed in the Source tab. A neat example of the product's habit: it would rather give you an approximate number labeled approximate than a precise-looking number that quietly isn't.

6.7 Company pipeline summary

The whole quarter in one card: \$518K booked of \$6.2M, \$5,198K open pipeline across 46 deals, and the \$484K uncovered gap. Two warnings worth reading aloud: 38% of pipeline is on recycled IDs, and — quietly the most strategic fact in the dataset — **the Prospecting stage is empty**. Nothing new is being seeded. Next quarter's pipeline problem is being created right now.

6.8 Data quality report ● Verified

Chapter 4 as a one-click answer. Green badge — the one thing this dataset supports with full confidence is a precise description of its own problems. Ends with the punchline: fix the source data and every flag in this tool clears itself automatically.

7 • The trust system

≈ 5 min

Trust in this product isn't a promise — it's five specific, visible mechanisms. Walk through them slowly; this is the chapter the CCO cares about.

1. The trust badge on every answer

BADGE	EXACT MEANING
● Verified	Computed from validated records and no known data problem touches this answer.
● Verified with assumptions	The arithmetic is exact, but a known data issue or a stated assumption affects how you should read it. The specifics are printed directly below.
● Insufficient data	A data defect <i>dominates</i> this answer — for example, most of the figure is re-dated revenue, or most of a rep's pipeline is on corrupted records. Treat it as a data problem to fix, not a conclusion to act on.

The badge is decided by fixed rules, not by feel — e.g. "if more than half of booked revenue is F4 re-dated, downgrade to red." The rules are in the methodology panel.

2. Source data on every answer

The actual rows. Not a sample — for these answer sizes, all of them. Tagged, so a row that carries a known issue says so right in the table ("ALSO in final Q1 revenue — close date moved").

3. The calculation, written out

Numerator, denominator, filter dates, thresholds — in the Calculation tab, legible enough to recompute on a napkin. Nothing is "proprietary scoring."

4. Warnings that attach themselves

Each of the five findings knows which answers it touches and prints itself there, quantified in dollars — not as a footnote in a manual somewhere.

5. Refusal as a feature

SHOW IN THE APP

Type: **What will Q3 revenue be?** Read the refusal out loud, including the list of what it *can* answer.

No Q3 data exists, so the tool says so and stops. The previous tool's fatal habit was improvising when it didn't know. This one treats "I can't answer that reliably" as a first-class answer.

The six standing assumptions

Also always on screen (sidebar → Methodology), briefly: the May 2 as-of date comes from the brief, not the files; quarter membership follows close dates; attainment counts only won deals; segment roll-ups use the deal's segment (verified consistent); open-deal close dates are forecasts, not facts; and "same point in the quarter" means equal days elapsed. Every answer lists which of these it used.

8 • The AI layer — what the model may and may not do

≈ 5 min

Expect this question in some form: *"so where exactly is the AI, and why won't it do what the last one did?"* The answer has three parts.

What the model is allowed to do — exactly two things

1. **Route:** when simple keyword matching can't place a question, the model picks which of the eight analyses fits — from a fixed list, returning a single choice that is then re-checked against the list. If it names a salesperson or a segment, that name is verified against the actual roster before use. Output that doesn't validate is discarded.
2. **Phrase:** rewrite an already-computed result into two to four friendly sentences — under the numeric veto described in Chapter 5. Every number in the sentence must exist, verbatim, in the computed result, or the whole sentence is discarded and replaced with the engine's own wording. When that happens, the tool says so on screen — the guardrail is visible, which is itself a trust feature.

What the model can never do

It cannot calculate, estimate, or round a number. It cannot write a database query. It cannot choose a filter, a date range, or a definition. It cannot see a question outside the menu and improvise. The numeric path — from the CSV files to the tiles on screen — contains no AI at any point.

Two properties worth saying out loud

- **It degrades gracefully.** No API key, network trouble, the vendor having an outage — the app quietly falls back to keyword routing and its own deterministic wording. We've had this happen live; the user experience is that summaries get a little plainer. Nothing breaks. Flip the sidebar toggle to demonstrate on demand.
- **It's vendor-agnostic.** During development this exact app ran on two different AI providers (Anthropic's Claude and Google's Gemini) — switched in one file, with *zero* change to any trust property. Because the guardrails live in our code, not in any vendor's model. That matters commercially: no lock-in, and the safety story survives procurement decisions.

KEY LINE — say it in these words

The last tool trusted a model to be right. This one assumes the model can be wrong, and builds the checking into the product. That's the whole difference — trust placed in verifiable code, with the AI kept on a short, visible leash for the two jobs it's genuinely good at.

9 • How everything was verified

≈ 4 min

"Can we trust it" deserves a concrete answer. Four layers:

- **38 automated tests.** Every important number — the \$518K, the \$1.26M, the 4.5%, each rep's risk verdict — was first computed by hand from the raw files, then locked into a test. If the data or a rule ever changes, a named test fails loudly. The tests also cover the edge rules (what happens exactly at a threshold) and the AI validator itself (it must reject a sentence containing \$999,999; it must reject "\$3.65M" as an abbreviation of \$3,650,000 — silent unit-changing is exactly the distortion we guard against).
- **5 reconciliations on every launch.** The same totals computed independent ways must agree — deal-level booked must equal per-rep booked must equal per-segment booked; every row must land in exactly one bucket. If any check fails, the app refuses to serve numbers at all. It fails closed, like a bank, not open, like a demo.
- **A measured router.** Routing accuracy isn't assumed — it's scored against a golden set of 24 phrasings: 21/24 on keyword rules alone, 22/24 with the AI fallback, and — the important part — **every miss fails to an honest refusal or a visibly-labeled interpretation**, never to a wrong number.
- **An adversarial audit.** After the tool worked, we attacked it: recomputed everything from raw files by hand looking for disagreements. That pass is what caught F4 — after all tests were green. The lesson is now product: F4 detection is automated, and the honest claim we make is not "never wrong" but "wrong is detectable, attributable, and cheap to fix."

KEY LINE — say it in these words

I'd rather show you the crack we found and how the system now catches it than claim there are no cracks. A tool that advertises its own limitations accurately is the only kind this organization should trust with numbers.

10 • What the business should do with this

≈ 4 min

The tool is the deliverable, but the data told a story with three action items. Present them as the analyst's summary — this is where you show business judgment, not just engineering.

1. **Create pipeline immediately.** The quarter's arithmetic doesn't close: \$484,000 uncovered even at a perfect win rate, coverage below the historical break-even in every segment, and — the leading indicator — **zero deals in Prospecting**. The Q1 finished-strong pattern (21% at day 32 → 102% final) buys some calm about the current pace, but nothing back-fills an empty top of funnel. This is the "act this week" item.
2. **Run a CRM hygiene sprint before the next QBR.** Twenty-eight deals carry defects; three reps can't even be evaluated because of them; and the F4 re-dating means Q1 and Q2 currently share \$373,000 of revenue. Fixing ~30 records is days of work, and every flag in this tool clears itself when they do. Owner: sales operations, with finance eyes on the F4 deals specifically.
3. **Adopt "strictly new" as a standing lens.** Until re-dating is impossible (see Chapter 11), leadership should look at both figures the tool shows: system-of-record bookings *and* strictly-new bookings. Today those are \$518K vs \$145K — different enough to change decisions.

And one meta-recommendation: make "what's wrong with my data?" a standing leadership question. The tool made it one click on purpose — data quality stopped being an IT topic the moment it started deciding which reps look like failures.

11 · Limits, and the road to production

≈ 3 min

What this prototype honestly doesn't prove

- **Scale:** it runs on 180 rows of CSV, not millions of rows of live CRM. (The design ports: the eight analyses become governed warehouse queries; the checks become pipeline tests.)
- **Open-ended language:** eight question types, honestly labeled — not "ask anything." Coverage grows intent by intent, each audited first.
- **Live messiness:** real Salesforce data has more failure modes than two snapshots can show. The audit posture — detect, disclose, downgrade — is the durable part.
- **No forecasting on purpose:** the data contains no stage-history from which honest win probabilities could be derived, so the tool refuses to weight its pipeline. Inventing probabilities is precisely how the last tool died; we'd rather show coverage against the real 77.9% win rate.

The production path, in one paragraph each

Weeks 1–3, foundation: the engine moves behind an API; CSVs are replaced by read-only connections to Salesforce data replicated into a warehouse (e.g. Snowflake); the validation checks move upstream into the data pipeline and block bad data before it's ever queryable; single-sign-on and role-based access (a regional leader sees their region; rep-level risk labels get governance with HR/sales ops before anyone sees them).

The one upgrade that matters most — snapshot journaling: recording every change to every deal, every day. With a change journal, F4-style re-dating isn't something a clever audit catches later — it's **impossible to hide** the moment it happens: "closed-won close dates don't move across quarter boundaries without a flag" becomes a standing monitored rule.

Then a pilot: three sales leaders, the current analyses, two commitments — every question the tool refuses becomes backlog for a new audited intent, and every challenged number gets an answer within a day. The pilot's success metric is behavioral: are leaders asking questions again? That was the original problem, and it's the thing worth measuring.

12 · Question-and-answer preparation

as needed

"How do we know it isn't hallucinating right now?"

Numbers never come from the model — they come from ordinary code you can read, tested against hand-checked values. The model's sentences pass a validator that rejects any number the code didn't produce, and you can see the check on screen ("11 numbers verified"). And here's the strongest answer: pick any number on this screen, and I'll show you the rows and the arithmetic behind it, live.

"Why can't I ask it anything I want?"

Because unlimited questions require the model to invent the analysis, and an invented analysis can be invisibly wrong — that's the exact failure you experienced. A short menu of audited analyses makes the worst case a visible misroute instead. The menu grows every week; the trust rule doesn't.

"Is Priya actually underperforming?"

Unknowable from this data, and the tool says so — 61% of her pipeline sits on corrupted records. The action is to fix her seven records, then re-ask. The tool distinguishing "bad data" from "bad performance" is the point.

"Which number is right — \$518K or \$145K?"

Both are computed correctly from different premises. \$518K is what the current system of record says Q2 has booked; \$145K excludes the three deals your closed Q1 book already counted. Which premise to bless is a finance-policy decision — the tool's job is to make sure you know there's a decision to make, which until now you didn't.

"What happens when the data gets fixed?"

Every finding is detected fresh from the files at each load. Fix the records, restart, and the flags disappear on their own — including the red badges. Nothing is hardcoded.

"Why did you build an app instead of a deck?"

Because the assignment's hard requirement — trust — can't be demonstrated on a slide. Refusals, red badges, and a validator have to be experienced to be believed. The deck version of "it shows source rows" is a claim; the app version is proof.

"What did it cost / what does it run on?"

It runs anywhere Python runs. AI usage is pennies per session — questions are routed for fractions of a cent and summaries are a few hundred words. And the AI is optional: with it off, the tool remains fully functional.

"Did you use AI to build it?"

Yes, as a pair-programmer — the brief encourages it and says how you use it is part of the evaluation. The division of labor: AI accelerated the code; the design decisions, the metric definitions, the audit that found F4, and the verification standards were mine — and every generated line ended up covered by the 38 tests. Using AI with guardrails to build a tool whose whole point is AI with guardrails felt appropriately consistent.

"What would you do with another four hours?"

Three things: a "challenge this number" button that files a review ticket with the answer's full context; region and product-line intents (the router already refuses those honestly); and exporting any answer — evidence, formula, flags — as a one-page PDF for meeting handouts.

Appendix A · Glossary — every term, in plain English

TERM	MEANING
Quota	A salesperson's dollar target for the quarter. Company total for Q2: \$6.2M across ten reps.
Closed-Won / booked	A deal that was actually won. The only thing that counts toward attainment.
Attainment	Booked revenue ÷ quota. "4.5%" means \$4.50 won for every \$100 of target.
Open pipeline	Total value of deals still being worked and expected to close this quarter. Potential, not revenue; never blended into attainment.
Coverage	Open pipeline ÷ the remaining gap to quota. 1.0× = you'd need to win literally everything. Below 1.0× = target arithmetically unreachable.
Break-even coverage (1.28×)	Coverage needed at the win rate Meridian actually achieved in Q1 — they won 77.9 cents of every decided dollar, and $1 \div 0.779 \approx 1.28$. Derived from their data, not an industry rule.
Pace	Attainment compared to how much of the quarter has elapsed (35% at day 32). Pace 1.0 = exactly on a straight-line track.
Same point in quarter	Equal days elapsed: Q2 through May 2 (day 32) vs Q1 through Feb 1 (its day 32).
Strictly new	Q2 bookings excluding the three F4 deals the closed Q1 book already counted. \$145K vs the headline \$518K.
Snapshot	A photograph of the whole deal table at a moment in time. We have two: quarter-end Q1, and May 2.
Hallucination	An AI stating something false with confidence — here specifically, inventing numbers. The failure this product is built to make impossible in the numeric path.
Deterministic	Same inputs, same outputs, every time — ordinary code, no AI involved. The property that makes checking possible.
Fail closed	When validation fails, show nothing rather than something possibly wrong.

Appendix B · Every number in the demo

Company, Q2 as of May 2

Quota (10 reps)	\$6,200,000	Booked	\$518,000 · 8.4%
Strictly new (excl. F4)	\$145,000	Open pipeline (46 deals)	\$5,198,000
Uncovered at 100% win	\$484,000	Quarter elapsed	day 32 of 91 · 35%

Q1 final (for context)

Won	\$6,041,000 · 51 deals · 102% of quota	Lost	\$1,711,000 · 12 deals
Value win rate	77.9% → 1.28× break-even	At its day 32	\$1,260,000 · 21.3%

Segments, Q2

SEGMENT	QUOTA	BOOKED	ATTAIN.	PIPELINE	COVERAGE	NOTE
Enterprise	\$3,650,000	\$165,000	4.5%	\$3,600,000	1.03×	booked is 100% F4; strictly new \$0
Mid-Market	\$2,040,000	\$353,000	17.3%	\$1,185,000	0.70×	\$208K of booked is F4
SMB	\$510,000	\$0	0%	\$413,000	0.81×	only segment with 0% recycled-ID pipeline

Reps, Q2 (booked · quota · coverage · verdict)

REP	BOOKED	QUOTA	COVERAGE	VERDICT
Sarah Chen (Ent, West)	\$165,000	\$950,000	0.86×	Watch — booked is 100% F4
Marcus Rivera (Ent, Central)	\$0	\$850,000	1.32×	Watch
Priya Patel (Ent, NE)	\$0	\$1,000,000	0.74×	Insufficient data — 61% recycled
James Okafor (Ent, SE)	\$0	\$850,000	1.25×	Watch
Danielle Torres (MM, West)	\$0	\$500,000	0.65×	Insufficient data — 54% recycled
Kevin Marsh (MM, NE)	\$0	\$540,000	0.74×	At risk
Aisha Williams (MM, Central)	\$60,000	\$480,000	0.57×	Insufficient data — 67% recycled
Tom Bradley (MM, SE)	\$293,000	\$520,000	0.97×	Watch — 56.3% attained; \$208K of it F4
Lisa Park (SMB, West)	\$0	\$270,000	0.71×	At risk
Ryan Cole (SMB, NE)	\$0	\$240,000	0.92×	At risk

Findings

WHAT	SCALE
F1 Recycled deal IDs (OPP-076...088)	13 deals · \$1.97M · 38% of pipeline
F2 Closed deals reopened/changed	7 deals · \$576K
F3 Open deals past forecast close	2 deals · \$87K
F4 Won in both quarters (dates moved) — OPP-042, 060, 074	3 deals · \$373K · 72% of Q2 booked
F5 Closed records edited after books closed (e.g. \$155K→\$120K)	3 deals · \$260K

Q2 open pipeline by stage

STAGE	DEALS	VALUE
Prospecting	0	\$0 — empty; nothing being seeded
Discovery	14	\$1,678,000
Proposal	19	\$1,977,000
Negotiation	13	\$1,543,000

Verification figures

38 automated tests · 5 load-time reconciliations · router scored 22/24 on a golden set (21/24 rules-only) · narration validator: 0 uncomputed numbers can render · adversarial audit caught F4 post-testing; detection now automated.

Appendix C · Suggested demo run-order

1. **Resting screen** (1 min) — four KPI cards; land on "\$484K uncovered."
2. **Enterprise vs quota** (3 min) — the red badge; the one-row numerator; strictly-new \$0.
3. **Reps at risk** (3 min) — coverage chart with the 1.0x/1.28x lines; three unclassifiable reps.
4. **Q2 vs Q1 pace** (2 min) — day-32 alignment; \$518K vs \$1,260K; strictly-new tile.
5. **Type "Are we going to make the number this quarter?"** (2 min) — LLM routing tag + "N numbers verified" line.
6. **How is Priya Patel doing?** (2 min) — the refusal to judge a person on corrupted records.
7. **Data quality** (2 min) — the green-badged audit; "fix the data and the flags clear themselves."
8. **What will Q3 revenue be?** (1 min) — the honest refusal. Close on: eight trusted answers beat eight thousand plausible ones.

Recovery moves: Gemini down → sidebar toggle off, narrate the graceful degradation as a feature. App down → present from the one-page leave-behind (docs/leave-behind.pdf) without breaking stride.

Prepared by Sai Saran · Meridian Systems sales-intelligence scenario · Companion artifacts: live prototype (run.sh), one-page summary (docs/leave-behind.pdf), this guide.